

SMART POLYPHARMACY AND VITAMIN RECOMMENDATION SYSTEM

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APRIL 2026

EARLY VITAMIN DEFICIENCY WARNING SYSTEM BASED ON SYMPTOMS AND LONG-TERM MEDICATION

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DECLARATION

I hereby declare that this dissertation is entirely my own work and has not been submitted, in whole or in part, to any other university or institute of higher learning for the award of any degree or diploma. To the best of my knowledge and belief, it does not contain any material previously published or written by another person, except where due acknowledgment has been made in the text.

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my dissertation supervisor, Ms. Jenny Krishara, and my co-supervisor, Dr. Dinuka R. Wijendra, for their invaluable guidance, continuous support, and constructive feedback throughout the course of this research. Their expertise, encouragement, and insightful suggestions played a vital role in shaping this study and improving its overall quality.

I am also grateful to the Faculty of Computing for providing the necessary academic support and resources during my research journey. The knowledge and guidance gained from the faculty members have been instrumental in the successful completion of this dissertation.

I would like to extend my appreciation to my teammates for their cooperation, discussions, and exchange of ideas, which contributed significantly to my understanding of the research domain.

Finally, I would like to thank everyone who supported me directly or indirectly throughout this research. Their encouragement and assistance have been greatly appreciated.

ABSTRACT

Early detection of vitamin deficiencies is critical for preventing long-term health complications, especially in individuals undergoing prolonged medication use. This study presents a machine learning–based approach for predicting potential vitamin deficiencies using textual data derived from drug interactions and reported symptoms. The proposed system formulates the problem as a multi-label classification task, recognizing that multiple deficiencies may occur simultaneously.

The dataset consists of medication pairs and symptom descriptions, which are preprocessed and transformed into numerical representations using Term Frequency–Inverse Document Frequency (TF-IDF). A baseline Logistic Regression model is initially developed to validate feasibility, followed by a Linear Support Vector Machine (SVM) for improved performance. To address the limitation of labeled data, a semi-supervised learning strategy is applied, where high-confidence predictions from the trained model are used to expand the dataset.

Additionally, label refinement is performed to ensure that only clinically relevant vitamin deficiency categories are retained, improving the semantic consistency of the model. Experimental results demonstrate strong predictive performance, particularly for commonly occurring deficiencies, with the final model achieving high weighted F1-scores.

The proposed system provides a scalable and interpretable solution for early vitamin deficiency warning, supporting healthcare decision-making by identifying potential risks based on medication and symptom patterns.

Keywords: vitamin deficiency prediction; multi-label classification; TF-IDF; Support Vector Machine; semi-supervised learning; drug interactions; symptom analysis.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
TF-IDF	Term Frequency–Inverse Document Frequency
SVM	Support Vector Machine
LR	Logistic Regression
OvR	One-Vs-Rest
EDA	Exploratory Data Analysis
CSV	Comma-Separated Values
API	Application Programming Interface
F1- score	Harmonic Mean of Precision and Recall
KNN	K-Nearest Neighbors
NB	Naïve Bayes
DL	Deep Learning
BERT	Bidirectional Encoder Representations from Transformers

1. INTRODUCTION

Early identification of vitamin deficiencies is essential for preventing long-term health complications and improving overall patient well-being. In modern healthcare settings, individuals are frequently exposed to prolonged medication usage, which can interfere with nutrient absorption and lead to deficiencies over time. These deficiencies often develop gradually and may present through subtle or overlapping symptoms, making early detection challenging through traditional clinical methods alone. As a result, there is a growing need for intelligent systems that can assist in identifying potential vitamin deficiencies at an early stage using available patient-related data.

However, predicting vitamin deficiencies is not a straightforward task. The relationship between medications, symptoms, and nutrient imbalances is complex and highly variable. A single medication can affect multiple vitamins, and different combinations of drugs may produce compounded effects. Additionally, symptoms such as fatigue, weakness, or neurological changes are often non-specific and may be associated with multiple deficiencies. Conventional approaches that rely solely on fixed medical rules or isolated symptom analysis may fail to capture these intricate relationships. Therefore, a more adaptive and data-driven approach is required to model these interactions effectively.

This research addresses this challenge by developing a machine learning-based system for predicting vitamin deficiencies using textual data derived from drug interactions and symptom descriptions. The system treats the problem as a multi-label classification task, recognizing that multiple deficiencies can occur simultaneously in a patient. To enable effective learning, medication names and symptom descriptions are combined and transformed into a structured numerical representation using Term Frequency–Inverse Document Frequency (TF-IDF), allowing the model to capture important textual patterns.

The proposed approach follows a progressive modeling strategy. Initially, a baseline Logistic Regression model is implemented to validate the feasibility of the task and establish a performance benchmark. Subsequently, a Linear Support Vector Machine (SVM) is introduced to improve classification performance, particularly in handling high-dimensional sparse text data. To address the limitation of labeled medical data, a semi-supervised learning strategy is incorporated, where the trained model is used to infer labels for previously unlabeled samples based on prediction confidence. This enables expansion of the training dataset while maintaining reliability.

Furthermore, to ensure the system remains aligned with the research objective, the label space is refined to include only clinically relevant vitamin deficiency categories. This step eliminates noise caused by unrelated medical annotations and improves the interpretability of the model outputs. The final system is designed to provide early warning signals for potential vitamin deficiencies based on medication and symptom patterns, supporting healthcare professionals in decision-making processes.

Overall, this research proposes a scalable, interpretable, and data-driven solution for early vitamin deficiency detection. By leveraging machine learning techniques and real-world medical text data, the system aims to enhance early diagnosis, reduce the risk of severe complications, and contribute to more proactive healthcare management.

1.1. Background Literature

The detection of vitamin deficiencies has traditionally relied on clinical evaluation, laboratory testing, and rule-based diagnostic systems. Early approaches, such as expert systems, utilized predefined medical rules to infer deficiencies based on symptoms. For example, systems based on forward chaining techniques demonstrated the ability to mimic clinical reasoning by mapping symptoms to possible vitamin deficiencies. While these systems provided a structured and interpretable approach, they were limited by their dependency on manually defined rules and their inability to adapt to complex or unseen patterns in data [3], [4].

With the advancement of artificial intelligence, machine learning techniques have been increasingly applied to medical diagnosis tasks, including vitamin deficiency detection. Neural network-based approaches, particularly Convolutional Neural Networks (CNNs), have been explored for classification tasks, demonstrating improved accuracy compared to traditional rule-based systems [1], [2]. These models can learn complex relationships between input features and output labels; however, they typically require large, well-annotated datasets and substantial computational resources. In many real-world medical scenarios, such large, labeled datasets are not readily available, which limits the practical applicability of deep learning approaches.

Several studies have also explored the use of classical machine learning algorithms for vitamin deficiency detection and recommendation systems. Multi-classifier approaches combining techniques such as decision trees, support vector machines, and naïve Bayes have shown promising results in predicting deficiencies and recommending dietary interventions [5]. These approaches are generally more interpretable and require less data compared to deep learning models, making them suitable for structured and semi-structured medical datasets.

From a clinical perspective, vitamin deficiencies such as Vitamin B12, Vitamin A, and Vitamin D have been extensively studied due to their significant health impacts. Vitamin B12 deficiency, for instance, is associated with neurological disorders and anemia, while Vitamin A deficiency affects vision and immune function, and Vitamin D deficiency has global prevalence with implications for bone health and chronic diseases [6], [7], [8]. These studies highlight the importance of early detection and continuous monitoring, particularly in populations exposed to risk factors such as poor nutrition or long-term medication use. Recent research has also explored the broader role of vitamins in preventing complex diseases, including neurodegenerative conditions, further emphasizing the need for predictive and preventive healthcare solutions [9], [10].

Despite these advancements, several limitations remain in existing systems. First, many approaches rely on structured clinical inputs or laboratory results, which may not always be readily available. Second, most systems are designed for single-label classification, assuming one deficiency per patient, whereas in reality multiple deficiencies can coexist. Third, limited attention has been given to integrating diverse textual sources such as drug interactions and symptom descriptions, which contain valuable contextual information for prediction.

Natural Language Processing (NLP) techniques offer a promising direction for addressing these challenges. Text-based representations, such as Term Frequency–Inverse Document Frequency (TF-IDF), enable the transformation of unstructured medical text into numerical features suitable for machine learning models. When combined with classifiers such as Logistic Regression and Support Vector Machines (SVM), these approaches have demonstrated strong performance in text classification tasks due to their ability to handle high-dimensional sparse data efficiently.

Furthermore, semi-supervised learning has emerged as an effective strategy for dealing with limited labeled data. By leveraging both labeled and unlabeled samples, models can expand their training data through high-confidence predictions, improving generalization without extensive manual annotation. This approach is particularly relevant in medical domains, where labeling data is expensive and time-consuming

Positioned within this context, the present study addresses the identified gaps by proposing a multi-label, text-based vitamin deficiency prediction system. Unlike traditional rule-based or single-label models, this approach integrates drug interaction data and symptom descriptions to capture complex relationships influencing vitamin deficiencies. The system employs TF-IDF for feature representation and compares classical machine learning models, selecting a Linear Support Vector Machine (SVM) as the final classifier. Additionally, a semi-supervised learning strategy is incorporated to enhance dataset size and model robustness. By focusing on interpretable, scalable methods and real-world textual data, this research contributes a practical solution for early vitamin deficiency detection aligned with current trends in data-driven healthcare.

1.2. Research Gap

Although significant progress has been made in the field of vitamin deficiency detection, several critical gaps remain when considering real-world applicability, particularly in early warning systems based on symptoms and long-term medication usage

First, many existing studies focus on **image-based detection techniques**, such as Convolutional Neural Networks (CNNs) applied to visual symptoms (e.g., skin, eyes, or tongue analysis) [1], [2]. While these approaches demonstrate high accuracy in controlled environments, they are limited in scope as they rely solely on visual indicators and fail to incorporate underlying causes such as medication usage or internal physiological changes. As shown in the comparison table, these methods do not support multi-vitamin detection or integration with long-term medication data.

Second, traditional **expert systems**, including those based on forward chaining, provide rule-based diagnosis using predefined symptom–vitamin relationships [3], [4]. Although interpretable, these systems lack adaptability and cannot handle complex, overlapping symptom patterns. They are typically restricted to single-deficiency predictions and do not support predictive risk scoring or learning from new data, which limits their effectiveness in dynamic healthcare environments

Third, prior research on **machine learning–based recommendation systems** has explored multi-classifier approaches for identifying deficiencies and suggesting dietary interventions [5]. While these systems improve prediction accuracy compared to rule-based methods, they often depend on structured inputs and do not fully utilize unstructured textual data such as combined drug interactions and symptom descriptions. Additionally, most of these systems do not incorporate long-term medication effects, which are a significant contributor to vitamin deficiencies.

Furthermore, a key limitation across most existing approaches is the **lack of multi-label prediction capability**. In real-world scenarios, patients may experience multiple vitamin deficiencies simultaneously. However, many systems are designed for single-label classification, resulting in incomplete or oversimplified diagnoses.

Another major gap is the **absence of symptom-based early warning systems integrated with medication history**. While some systems consider symptoms or provide mobile-based accessibility, very few combine both symptom analysis and medication-based risk factors into a unified predictive framework. As illustrated in the comparison table, none of the existing approaches fully support multi-factor health data integration, including diet, lifestyle, and medication.

In addition, **adaptive learning and scalability** remain underexplored. Most prior systems are static and do not improve over time with new data. They lack mechanisms such as semi-supervised learning, which can leverage unlabeled data to enhance model performance—an important requirement in medical domains where labeled data is scarce.

Finally, there is limited focus on **non-image textual data processing**, despite the availability of rich information in clinical notes, drug interactions, and symptom descriptions. Existing systems largely ignore this data modality, missing an opportunity to build more comprehensive and accessible prediction models.

Resulting gap. There is no unified approach that (i) integrates long-term medication data with symptom-based analysis, (ii) supports multi-label prediction of vitamin deficiencies, (iii) leverages unstructured textual data using machine learning techniques, and (iv) incorporates adaptive learning to improve performance over time. Addressing this gap is essential for developing a practical, scalable, and early warning system capable of assisting healthcare professionals in identifying potential vitamin deficiencies before they progress into severe conditions.

Feature / Aspect	Research [1](CNN Image Detection)	Research [2](Neural Network Image Based)	Research [3](Expert System Forward Chaining)	Research [4](Expert System Mobile Based)	Proposed Solution (Symptom + Medication-Based Warning)
Multi-vitamin detection	No	No	No	No	Yes
Integration with long term medication	No	No	No	No	Yes
Symptom-based early warning	No	No	Yes	Yes	Yes
Predictive risk scoring	No	No	No	No	Yes

Personalized preventive Guidance	No	No	No	Yes	Yes
AI / ML-driven analysis	Yes	Yes	No	Partially	Yes
Mobile / web accessibility	No	No	No	Yes	Yes
Adaptive system learning	No	No	No	No	Yes
Non-image input (symptoms & history)	No	No	No	No	Yes
Multi-factor health data (diet, lifestyle, medication)	No	No	No	No	Yes

Table 1: Comparison between existing systems

1.3. Research Problem

The problem addressed in this study is the development of an adaptive, data-driven system for early prediction of vitamin deficiencies based on drug usage patterns and symptom descriptions. In real-world healthcare scenarios, individuals often experience multiple overlapping vitamin deficiencies that develop gradually due to long-term medication use, poor absorption, or drug–nutrient interactions. However, identifying these deficiencies at an early stage remains challenging due to the complexity and non-specific nature of symptoms.

Traditional diagnostic approaches rely heavily on clinical testing or rule-based expert systems, which are limited in their ability to generalize across diverse patient conditions. These methods typically assume a single deficiency per case and depend on manually defined relationships between symptoms and vitamins. As a result, they are unable to effectively model complex, multi-label relationships or adapt to unseen patterns in real-world data.

In this project, the input consists of structured medication pairs (Drug1 and Drug2) combined with unstructured symptom descriptions. The core problem is to learn a mapping from this combined textual representation to one or more vitamin deficiency labels. Since multiple deficiencies can co-exist, the task is formulated as a multi-label classification problem rather than a single-label prediction task.

The proposed system must also operate under practical data constraints. Medical datasets are often incomplete, with missing labels and limited annotated examples. Therefore, the model must be capable of learning effectively from small, labeled datasets while also leveraging unlabeled data where possible. Additionally, the system must handle inconsistent and noisy vitamin label representations, ensuring that predictions remain clinically meaningful and standardized.

Formally, the research problem can be defined as follows:

1. Given a combined textual input consisting of Drug1, Drug2, and symptom descriptions, predict a set of vitamin deficiency labels that may co-exist for a single case.
2. Learn a robust mapping function that:
 - captures relationships between medications and symptoms
 - supports multi-label outputs
 - handles imbalanced and sparse labeled data
 - remains interpretable for healthcare use cases
3. Improve prediction performance using realistic constraints, including:
 - limited labeled medical data
 - noisy and inconsistent label formats
 - reliance on text-only clinical information without laboratory values
4. Ensure that the system can be extended using semi-supervised learning, where high-confidence predictions on unlabeled data can be incorporated to improve model generalization.

Highly unbalanced vitamin label distribution, inconsistent clinical annotations, overlapping symptom-deficiency correlations, and the difficulty of modelling multi-label dependencies from unstructured text data are some of the major issues that characterize this research subject. The process is made more challenging by the need to ensure generalization across unknown medication combinations and symptoms.

By combining traditional multi-label machine learning models, TF-IDF-based text representation, and a semi-supervised learning approach to increase training data, the work tackles these issues. The anticipated result is a trustworthy early warning system that may help proactive healthcare decision-making by forecasting possible vitamin deficiencies using drug and symptom data.

1.4. Research Objectives

This study's primary goal is to create an adaptive machine learning-based system that uses medication combinations and symptom descriptions to identify vitamin deficiencies early on, allowing for precise multi-label categorization for medical decision support. The system's objectives are to interpret clinical symptoms and pharmaceutical data as structured text inputs, use suitable feature extraction methods like TF-IDF, and discover connections between medications and deficient results. In order to forecast several potential vitamin deficiencies for a single patient, it also focuses on developing and accessing reliable classification models like Support Vector Machines and Logistic Regression. Furthermore, by utilizing high-confidence predictions from unlabeled data, the study uses a semi-supervised learning strategy to enhance model performance while maintaining the system's generalizability, interpretability, and applicability to actual clinical situations..

1.4.1. Main Objectives

The primary goal of this research is to create an intelligent machine learning-based system that can accurately classify multiple labels by combining medication combinations and symptom descriptions to diagnose vitamin shortages early. In order to improve performance through semi-supervised learning by utilizing high-confidence predictions from unlabeled data, the system seeks to learn significant relationships between pharmaceuticals and clinical symptoms from text data. All the while maintaining the model's accuracy, scalability, and suitability for real-world healthcare decision support.

1.4.2. Sub-Objectives

- 2.** Patient data collection → self-reported health profile.
 - o Build a structured framework to collect patient-reported data such as symptoms (fatigue, weakness, mood changes, sleep issues), diet patterns, lifestyle habits, and long-term medication history to form the core dataset for analysis.
- 3.** Data structuring & preprocessing → clean health input pipeline.
 - o Standardize and preprocess incoming patient data to handle missing values, normalize symptom scales, and convert free-text inputs into structured features suitable for AI modeling.
- 4.** AI-based deficiency risk analysis → pattern detection engine.
 - o Apply machine learning models to identify patterns between symptoms, diet, lifestyle, and medication use to estimate the likelihood of vitamin deficiencies such as B12, D, and folate, including risk severity scoring.
- 5.** Personalized recommendation system → tailored intervention output.
 - o Generate customized dietary plans, supplementation suggestions, and lifestyle adjustments based on predicted deficiencies, patient age, health conditions, and medication interactions.
- 6.** Real-time alert system → high-risk notification layer.
 - o Implement threshold-based and AI-driven alerts to notify users when symptoms or patterns indicate high deficiency risk, prompting early action or medical consultation.
- 7.** Longitudinal monitoring → continuous health tracking.
 - o Track patient symptoms, dietary intake, and lifestyle changes over time to observe trends, measure improvement, and continuously refine risk predictions and recommendations.
- 8.** User engagement interface → patient-friendly interaction layer.
 - o Design an intuitive platform for daily symptom logging, diet tracking, and feedback on recommendations to ensure consistent usage and long-term adherence.
- 9.** Clinical decision support → doctor-facing intelligence dashboard.
 - o Provide summarized reports, risk analytics, and trend visualizations to healthcare professionals to support evidence-based decisions for testing, diagnosis, and treatment planning.
- 10.** Model improvement loop → adaptive learning system.
 - o Continuously refine AI models using new patient data and feedback from outcomes to improve prediction accuracy and personalization over time.
- 11.** Safety, privacy & compliance → secure health data handling.
 - o Ensure secure storage, encryption, and privacy-aware processing of patient data while maintaining compliance with relevant healthcare data protection standards.

- 12.**Evaluation metrics & validation → system performance tracking.
- o Measure system effectiveness using metrics such as prediction accuracy, early detection rate, user adherence, and improvement in patient vitamin levels after interventions.

13.Methodology

13.1. Methodology

13.1.1. System Overview

Through ongoing analysis of patient-reported symptoms, lifestyle behavior, food habits, and long-term prescription usage, this method is intended to facilitate early diagnosis of possible vitamin deficiencies. It functions as a closed-loop health intelligence system that gathers self-reported data, transforms it into structured characteristics, uses AI models to assess deficiency risk, and constantly improves predictions based on user input and long-term monitoring. The primary objective is to detect potential deficiencies such as vitamin B12, vitamin D, and folate early on and provide prompt action before circumstances worsen.

Patients use a digital interface to enter structured and semi-structured data during the data entry step. This covers symptoms like weakness, exhaustion, mood swings, and trouble sleeping, as well as eating patterns, supplement consumption, and past pharmaceutical use. Preprocessing methods like normalization, encoding, and imputation are used to manage missing or inconsistent data when these inputs are standardized into a uniform feature format. This guarantees the data's dependability and suitability for AI-based analysis.

After processing, the system uses machine learning models trained on the correlations between symptoms, diet, and medications to assess the patient profile. These algorithms determine a risk level for each ailment and assess the probability of certain vitamin deficiencies by analyzing patterns across several dimensions. The result is a multi-risk profile that shows which vitamins may be deficient and how serious the danger is. Both consumers and healthcare professionals can comprehend why a particular insufficiency is feared thanks to a straightforward interpretability layer that guarantees forecasts are explicable.

A recommendation engine creates tailored advice based on the anticipated hazards. This includes lifestyle changes, food advice, and supplement recommendations based on the patient's age, medical problems, and drug interactions. The method makes sure that recommendations don't contradict current medical treatments or result in needless supplementing by giving priority to safe and practical initiatives.

A real-time alert mechanism is integrated to trigger notifications when symptom patterns or model outputs indicate a high probability of deficiency. These alerts are designed to encourage early medical consultation and prevent progression to severe deficiency states. The system ensures that high-risk users are flagged immediately while lower-risk cases are monitored continuously without causing fatigue alert.

By continually monitoring changes in symptoms, food, and medication use over time, the platform also facilitates longitudinal monitoring. This enables the system to assess the effectiveness of actions and the improvement or worsening of risk levels. The patient profile is updated with each new data entry, allowing for dynamic recalibration of forecasts and suggestions.

For healthcare providers, the system generates structured summary reports that include risk scores, symptom trends, dietary gaps, and medication interactions. This supports clinical decision-making by providing an evidence-based overview that can guide further testing, diagnosis, or treatment planning.

Overall, the system operates as a continuous learning loop where patient input drives prediction, prediction drives intervention, and outcomes feed back into improved future accuracy. The focus is on early detection, personalized intervention, and long-term health tracking, ensuring a proactive approach to managing vitamin deficiencies rather than reactive treatment.

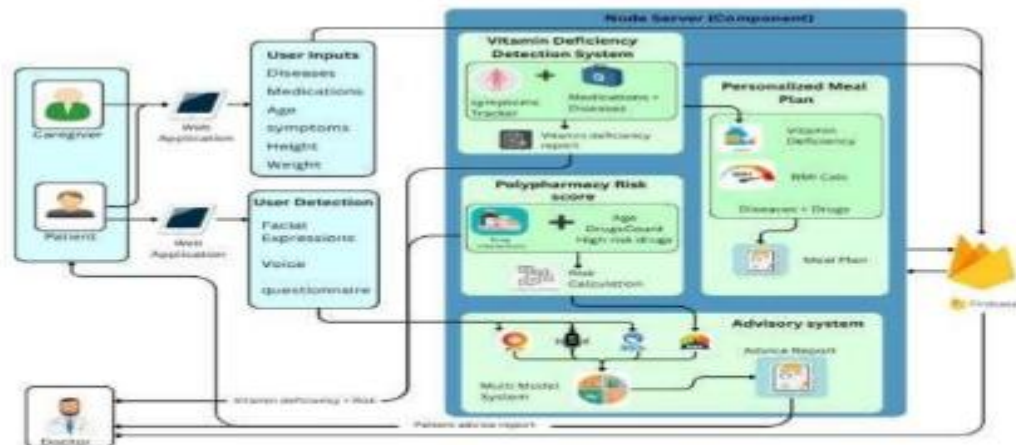


Figure 2: the system overview diagram

Figure 1: System Overview Diagram

13.1.2. Data Collection and Preprocessing

Because there is no publicly available dataset that captures patient-reported symptoms, lifestyle behavior, dietary intake, and long-term medication history in a structured format suitable for vitamin deficiency prediction, the dataset for this system is built using a hybrid approach of real-world self-reported inputs and structured synthetic augmentation. This is necessary to ensure the model can generalize across diverse populations, symptom variations, and medication combinations, rather than being limited to a narrow clinical dataset.

Patient-reported dataset for generalization

Why hybrid data collection is used: Real clinical datasets for vitamin deficiency are often incomplete, unbalanced, or unavailable at symptom-level granularity. Patients rarely report standardized symptom logs, and medication history is usually fragmented. Therefore, a structured data collection framework is designed to unify real user inputs with controlled synthetic cases. This allows the system to learn from both realistic behavior patterns and edge-case medical scenarios such as long-term deficiency progression or conflicting medication effects.

What each sample contains:

Each patient record is structured into a unified feature vector representing:

- **Symptom profile:** fatigue, weakness, dizziness, mood changes, sleep disturbances, hair loss, and cognitive fog (encoded with severity scores)
- **Dietary intake:** frequency of vitamin-rich foods (B12, D, folate sources), meal consistency, and dietary restrictions
- **Lifestyle factors:** sunlight exposure, physical activity level, stress level, and sleep duration
- **Medication history:** long-term drug usage (e.g., antacids, metformin, anticonvulsants) that may interfere with nutrient absorption
- **Demographic context:** age group, gender, and general health condition category
- **Output label (training phase only):** likelihood of vitamin deficiencies such as B12, D, or folate with severity level

Each record is stored as a structured vector with normalized scales to ensure consistency across different input types.

Scenario coverage:

To ensure robustness, the dataset generation and collection framework covers a wide range of realistic and medically relevant scenarios:

- **Symptom variability:** mild, moderate, and severe fatigue patterns; short-term vs long-term weakness; fluctuating mood and sleep disorders
- **Diet patterns:** vegetarian, vegan, mixed diet, and nutrient-deficient diets with irregular meal timing
- **Medication interactions:** long-term use of drugs known to reduce vitamin absorption (e.g., metformin → B12 risk, PPIs → magnesium/B12 impact)
- **Lifestyle differences:** high vs low sunlight exposure (urban vs rural behavior), sedentary vs active individuals
- **Progression stages:** early deficiency signals (subtle symptoms) to advanced deficiency patterns (multiple symptom co-occurrence)
- **Population diversity:** different age groups (young adults, middle-aged, elderly) with varying absorption efficiency and risk profiles

Data structuring and storage:

Each patient entry is converted into a standardized feature schema before storage:

- All symptoms are mapped into numerical severity scores (0–5 scale)
- Dietary intake is converted into frequency-based nutrient indicators
- Medication history is encoded using categorical embeddings for drug classes
- Lifestyle factors are normalized into risk-impact scores
- Final dataset is stored in structured format (e.g., CSV/JSON or database tables) for model training

This structured representation ensures compatibility with machine learning models while preserving interpretability for clinical use.

Expert validation:

All synthetic and rule-based mappings are validated with guidance from healthcare domain knowledge to ensure medical plausibility. For example:

- Chronic fatigue + poor diet + metformin usage → increased B12 deficiency risk
- Low sunlight exposure + limited dietary intake → Vitamin D deficiency likelihood
- Mood changes + sleep disturbances + folate-poor diet → potential folate deficiency

Invalid or medically unrealistic combinations are removed or corrected during dataset refinement.

Dataset hygiene and quality control:

To ensure reliability of the dataset, strict validation rules are enforced:

- Standardized feature format across all records
- Removal of incomplete or contradictory entries
- Handling of missing values using medically informed imputation strategies
- Detection of outliers in symptom severity or dietary patterns
- Continuous logging of dataset distribution (to prevent class imbalance issues in deficiency types)

Any failing record validation is discarded rather than forced into the dataset to maintain clinical reliability.

Normalization policy:

All inputs are normalized before model training while preserving interpretability:

- Symptom severity → scaled (0 to 1 or 0 to 5 normalized range)
- Dietary intake → frequency per week normalized
- Medication duration → standardized in months/years
- Lifestyle factors → risk-index scaled values

Raw values are still retained in logs to allow traceability and clinical explanation during result interpretation.

13.1.3. Functional, Non-Functional Requirements**Functional Requirements****1. Patient data collection and symptom logging**

1.1 The system shall collect patient-reported data at regular intervals through a digital interface, including symptoms, diet, lifestyle habits, and medication history.

1.2 The system shall allow structured input of symptoms such as fatigue, weakness, mood changes, sleep disturbances, and cognitive issues using standardized severity scales.

1.3 The system shall store all patient inputs in a structured format for further processing and analysis.

Acceptance: Each patient entry is successfully recorded with complete feature fields or flagged if incomplete.

2. Data preprocessing and standardization

2.1 The system shall convert raw patient inputs into a standardized feature vector suitable for machine learning processing.

2.2 The system shall normalize symptom severity, dietary frequency, medication duration, and lifestyle indicators into consistent numeric scales..

2.3 The system shall handle missing or inconsistent values using predefined imputation rules or rejection criteria.

Acceptance The system shall handle missing or inconsistent values using predefined imputation rules or rejection criteria

3. Mode-based deficiency classification

3.1 The system shall classify each case into deficiency risk categories (low, moderate, high) based on model output thresholds.

3.2 The classification shall be updated at each new patient data submission without relying on previous stale states.

3.3 Acceptance: Each update produces a refreshed risk classification per vitamin type.

4. Duration-style risk progression estimation

4.1 The system shall estimate a risk progression indicator representing how long symptoms have been developing based on medication history and symptom trends.

4.2 The system shall convert this progression indicator into a normalized feature for use in downstream prediction models.

Acceptance: A valid progression value is generated for each patient profile update..

6. Personalized recommendation generation

6.1 The system shall generate personalized dietary recommendations based on detected vitamin deficiency risks.

6.2 The system shall provide supplementation suggestions and lifestyle modifications tailored to age, health status, and medication interactions.

6.3 The system shall ensure recommendations remain medically safe and avoid conflicts with existing medications.

Acceptance: Each high or moderate risk case produces at least one actionable recommendation set.

7. Alert and notification system

7.1 The system shall generate alerts when patient risk levels exceed predefined thresholds for any vitamin deficiency.

7.2 The system shall notify users in real time when high-risk patterns are detected.

7.3 The system shall escalate repeated or severe risk cases as priority alerts recommending medical consultation.

Acceptance: High-risk conditions trigger notifications within the same processing cycle.

8. Longitudinal monitoring and tracking

8.1 The system shall continuously track patient symptoms, dietary habits, and medication usage over time.

8.2 The system shall update risk predictions dynamically based on newly entered historical data.

8.3 The system shall maintain a timeline of patient health profiles for trend analysis.

Acceptance: System retains multi-session history and updates risk trends per entry.

9. Clinical decision support output

9.1 The system shall generate structured summary reports of patient risk profiles for healthcare providers.

9.2 The system shall include symptom trends, dietary gaps, medication interactions, and predicted deficiency risks in reports.

9.3 The system shall present outputs in an interpretable format suitable for clinical

review.

Acceptance: Each patient profile can be exported as a structured clinical summary.

10. Safety, validation, and fallback handling

10.1 The system shall validate all inputs to ensure they are within medically plausible ranges.

10.2 The system shall reject or flag inconsistent or incomplete patient entries instead of forcing predictions.

10.3 If AI prediction is unavailable, the system shall fall back to rule-based risk estimation.

Acceptance: No invalid or corrupted data is used for final prediction output.

11. System performance and real-time operation

11.1 The system shall process patient input and generate risk predictions in real time.

11.2 The system shall ensure that inference latency remains suitable for interactive use in a clinical or mobile environment.

Acceptance: Prediction cycle completes within acceptable response time for user interaction.

12. Privacy and data security compliance

12.1 The system shall securely store all patient data using encryption mechanisms.

12.2 The system shall ensure that only authorized access is allowed for patient records and clinical reports.

12.3 The system shall maintain data integrity across all processing stages.

Acceptance: All patient data is protected and accessible only through authenticated access layers.

3. Vitamin deficiency risk analysis (AI model)

3.1 The system shall analyze processed patient data using machine learning models to estimate the risk of vitamin deficiencies such as B12, D, and folate.

3.2 The system shall detect patterns between symptoms, diet, lifestyle, and medication usage that indicate potential deficiencies.

3.3 The system shall output a risk score and severity level for each suspected vitamin deficiency.

Acceptance: Risk prediction is generated for each vitamin category with a numeric confidence score.

Non-Functional Requirements

1. Accuracy and effectiveness

- The system shall accurately detect vitamin deficiency risks from symptoms, diet, and medication history, with minimal false predictions. It shall provide consistent risk scores and reliable personalized recommendations across different users and conditions.

2. Reliability and robustness

- The system shall continue functioning even with missing or incomplete data using fallback methods. It shall handle noisy or partial inputs without failure and recover automatically when full data becomes available.

3. Safety and compliance

- The system shall follow medical guidelines for nutrition and avoid unsafe recommendations. It shall ensure all outputs are safe, non-diagnostic, and suitable for general health guidance only.

4. Portability and deployment

- The system shall run on web and mobile platforms with low computing requirements. The trained models shall be easily deployable on standard devices and scalable for future expansion.

5. Explainability

- The system shall clearly explain risk predictions by linking symptoms, diet, and medication history to the output. It shall provide simple traceability from input data to final recommendation

2.2. Commercialization Aspects of the Product

The suggested system is a software-based health intelligence layer that predicts early vitamin deficiency risks, including vitamin B12, D, and folate deficits, by analyzing patient-reported symptoms, food, lifestyle, and long-term prescription history. In order to stop deficits before they become clinically serious, it is intended to offer early warnings, individualized health advice, and ongoing monitoring. Healthcare professionals, clinics, wellness platforms, and people with chronic illnesses that want early, data-driven insights are the main users.

The system may be linked as an API/SDK into already-existing healthcare or wellness applications, or it can be offered as a mobile or web-based AI platform. It operates utilizing common user inputs via forms or digital health records and doesn't require any extra gear. To protect patient privacy, all processing is done locally or securely in the cloud. If AI models or external data (such as weather or medicine APIs) are not accessible, it may safely degrade to rule-based risk assessment.

A risk scoring engine that converts outputs into clinically meaningful deficiency levels, a multi-source symptom-diet-medication AI model trained on structured and synthetic health profiles, and a personalized recommendation system that offers safe dietary and supplementation guidance based on medical constraints are the three primary sources of differentiation. Additionally, the system is built to provide longitudinal monitoring, which enables forecasts to be continuously improved over time.

The symptom-to-risk mapping model, data preprocessing and standardization pipeline, medication-nutrient interaction logic, and the personalized recommendation engine in conjunction with temporal health tracking are examples of protectable intellectual property. In order to make forecasts visible for both consumers and medical experts, the system also incorporates explainable AI techniques.

Simulation-based model training and validation utilizing structured datasets is the first step in a workable go-to-market plan. Pilot deployment in clinics or wellness platforms comes next. Subscription-based SaaS solutions for healthcare providers and API licensing for digital health apps follow. Early detection accuracy, a decrease in late-stage deficiency instances, and user interaction with suggestions are important performance indicators.

Data validation rules, backup rule-based inference, and ongoing model retraining with fresh data all help to reduce risks including inadequate patient data, self-reporting bias, and model misclassification. Clinical usefulness and trust are guaranteed by adherence to safe recommendation bounds and healthcare data privacy regulations. In order to assist preventative medicine and enhance long-term patient health outcomes, the system combines AI-driven prediction with individualized healthcare recommendations to offer a non-invasive, scalable, and intelligent solution for early vitamin insufficiency identification.

2.3. Testing and Implementations

2.3.1. Implementations

2.3.1.1. AI Based Vitamin deficiency Risk Environment Development

Each patient is represented by the system as a structured health status vector based on long-term medication history, dietary habits, lifestyle variables, and symptom reports. The environment generates a multi-channel observation tensor at each evaluation step that includes: (i) normalized symptom severity profile (e.g., fatigue, weakness, mood swings, sleep problems, etc.); (ii) dietary intake and nutrient availability indicators (e.g., sources of vitamin B12, D, and folate); and (iii) normalized medication interaction features that represent medications that may impact nutrient absorption.

The model's output includes a continuous risk score and several vitamin risk projections (B12, D, and folate), each of which is translated into discrete risk categories (low, moderate, and high). The "action space" of the system is represented by these outputs, where each prediction is associated with a deficiency probability and severity categorization for each vitamin category.

Health dynamics are simulated by updating patient risk states based on new symptom inputs and lifestyle changes over time. Two core safety and logic guardrails are embedded in the environment:

- **Mode recomputation each step:** The system recalculates risk levels at every new data entry, ensuring that predictions reflect the most recent patient condition and preventing outdated risk states from influencing decisions.
- **High-risk override logic:** If severe symptom clusters or long-term medication interactions are detected (e.g., chronic fatigue + poor diet + absorption-affecting drugs), the system automatically escalates the risk level for the most affected vitamin category to a higher priority state to ensure early warning activation.

Prediction accuracy determines rewards (during training): accurate deficiency risk categorization is rewarded with positive reinforcement, but misclassification or significant divergence from ground-truth labels is penalized in proportion to the severity of the mistake. Better congruence with actual clinical trends is ensured by doing this. Episodes end either after a predetermined number of data cycles or when prediction stability is reached (consistent risk categorization across successive updates). A carefully selected collection of actual patient inputs and structured synthetic instances created to address uncommon deficient conditions are used to train the environment.

A temporal health model is used to calculate the progression feature, which predicts the length of deficiency-related symptoms based on treatment use and symptom history. The raw number in weeks or months is kept for clinical interpretation and assessment, although this value is normalized for model input stability.

2.3.1.2. Decision-Making Function Development

Based on the patient scenario, a decision function is used to evaluate the risk of vitamin deficiencies and establish the appropriate response level. In addition to fundamental contextual criteria like age group and health condition category, it evaluates total nutritional risk status using patient-reported symptoms, food habits, lifestyle behavior, and medication history. The system then uses this data to create a standardized risk threshold framework for vitamin B12, D, and folate deficits that is in line with clinical nutritional knowledge.

For every vitamin category, the desired result is a structured risk level (low, moderate, or high) that is produced using learned model patterns in conjunction with predetermined health and nutrition requirements. To prevent overestimation from isolated symptoms, safety limitations are incorporated to guarantee that risk escalation only happens when many associated signs (such as weariness + poor diet + long-term medicine usage) are present.

The system uses rule-based fallback logic utilizing existing symptom patterns and demographic context to estimate baseline risk level in the event that entire patient data is not available (e.g., missing dietary or medication history). This guarantees uninterrupted operation even in situations with incomplete data.

The AI recommendation module receives the final decision output, which is a scalar risk score and matching categorization for each kind of vitamin, in order to provide clinical warnings, dietary recommendations, and supplement recommendations. In order to guarantee safe and understandable decision-making in practical applications, the framework also conforms to standard nutritional and preventive healthcare principles.

2.3.1.3. AI Model development (Risk Prediction System)

State (S). A structured patient feature tensor consisting of symptom profile (fatigue, weakness, mood changes, sleep issues), dietary intake indicators (Vitamin B12, D, folate sources), lifestyle factors, and long-term medication history, all normalized into a unified numerical representation.

Action (A). A multi-output risk prediction vector for Vitamin B12, Vitamin D, and folate, where each output is classified into discrete risk levels (low, moderate, high) along with a continuous risk score (0–1).

Policy (π). A deep neural network with convolutional and fully connected layers that learns relationships between symptoms, diet, and medication interactions, producing vitamin-specific risk predictions for each patient state.

Value function (V). A parallel regression head that estimates overall deficiency likelihood confidence, used to stabilize training and improve prediction reliability across uncertain or incomplete inputs.

Reward (R). Training feedback is based on prediction accuracy against labeled or validated health outcomes, where correct risk classification receives positive reinforcement and incorrect or over/under-estimation is penalized proportionally to severity.

Training follows a supervised-to-reinforcement hybrid approach, where the model is first pre-trained on structured patient datasets to learn baseline symptom–nutrition relationships. This is followed by fine-tuning using reinforcement-style optimization to improve robustness under noisy, incomplete, or real-world patient inputs. Additional regularization techniques such as class balancing and entropy control are applied to prevent bias toward dominant deficiency classes.

To improve clinical reliability, the system is also initialized using rule-guided priors derived from known medical relationships (e.g., metformin → B12 risk increase, low sun exposure → Vitamin D deficiency). This ensures early-stage predictions are medically grounded before full data-driven learning takes over.

2.3.1.4. Multi-Agent Approach

To improve prediction stability across different severity levels of deficiency risk, a dual-model setup is adopted. **Agent-Low** handles low to moderate risk cases where symptoms are subtle and early-stage (e.g., mild fatigue, slight dietary gaps). **Agent-High** handles moderate to high risk cases where multiple symptoms and long-term medication interactions indicate stronger deficiency likelihood.

At runtime, the controller selects the appropriate agent based on the overall risk score or dominant symptom pattern. This specialization improves classification consistency, reduces prediction ambiguity, and enhances sensitivity to advanced deficiency stages without changing the input format or system interface.

2.3.1.5. Risk Aggregation and Recommendation Output

The system converts individual symptom-level and nutrient-level predictions into a unified patient risk profile for each vitamin category (B12, D, folate). Since recommendations cannot be generated per micro-feature independently, all feature-level outputs are aggregated into a single interpretable risk score per vitamin.

For each vitamin type, the system combines symptom severity signals, dietary insufficiency indicators, and medication interaction effects to produce a final risk classification (low, moderate, high). If multiple high-impact signals are detected, the system prioritizes the highest contributing factor to ensure clinically meaningful output.

The final outputs are structured into a single recommendation layer, where dietary guidance, supplementation advice, and lifestyle suggestions are generated per vitamin deficiency category. A shared confidence score is applied to all recommendations to indicate prediction reliability.

All outputs are rate-limited and normalized to ensure stability across repeated updates. The system assumes a practical deployment context where one unified patient profile is processed at a time, ensuring compatibility with real-world healthcare dashboards, mobile applications, and clinical decision support tools.

2.3.2. Testing

This section verifies that the proposed system accurately detects vitamin deficiency risks, generates safe and meaningful recommendations, and maintains stable performance under real-world variability in patient-reported data. Testing is structured from module-level validation to full end-to-end system evaluation, including stress testing, robustness checks, and clinical plausibility assessment. All tests are reproducible using the structured patient dataset combining real inputs and simulated health profiles, along with the trained AI models exported in deployable formats (e.g., H5/ONNX).

At the module level, each component is independently tested, including symptom processing, dietary feature extraction, medication encoding, and risk prediction. Every component, including the risk prediction, drug encoding, dietary feature extraction, and symptom processing modules, is examined separately at the module level. These tests guarantee proper standardization of input data, suitable handling of missing values, and consistency of results over several runs. Additionally, unit tests confirm that nutritional risk indicators and symptom severity mapping adhere to standard medical reasoning.

Integration tests evaluate the k modules for prediction. These tests guarantee proper standardization of input data, suitable handling of missing values, and consistency of results over several runs. Additionally, unit tests confirm that nutritional risk indicators and symptom severity mapping adhere to standard medical reasoning.

From raw patient input to the final categorization of vitamin deficiency risk and suggestion generation, the whole pipeline is assessed at the integration level. This guarantees that interactions between symptoms, food, and medicine are accurately recorded and represented in the finished product. When inputs are changed over time, the system is evaluated to ensure consistency across various patient profiles and reliable predictions.

The system's forecast accuracy, reaction speed, and stability under constant updates are assessed during performance testing. It has been confirmed that risk forecasts and suggestions are produced with a latency that is suitable for real-time or almost real-time applications. To make sure it can manage several patient profiles without experiencing performance deterioration, the system is also evaluated for scalability.

Extreme and edge-case situations, such as severely unbalanced diets, long-term pharmaceutical use, missing data inputs, and contradictory symptom reports, are used in stress testing. These tests make that the model can smoothly degrade utilizing fallback logic as needed and does not provide unstable or dangerous outputs under unexpected circumstances.

Robustness testing assesses the system's performance when patient inputs are inconsistent, loud, or incomplete. It guarantees that final risk projections are not much impacted by modest missing data or inaccurate submissions. Additionally, sensitivity analysis is used to see how minor dietary, or symptom changes impact the estimated levels of insufficiency risk.

Compliance and safety testing ensures that all generated recommendations remain aligned with nutritional safety guidelines and do not suggest harmful

supplementation or incorrect medical conclusions. The system is validated to ensure it operates as a decision-support tool rather than a diagnostic engine.

Lastly, user-level validation is carried out to evaluate the results' interpretability and usefulness. This involves determining whether risk categories, contributing variables, and suggested measures are easily understood by patients and healthcare professionals. These assessments provide feedback that is utilised to enhance overall system dependability and model behaviour.

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2.3.2.1. Test objectives and scope

The testing process aims to ensure the system correctly processes patient inputs and produces accurate vitamin deficiency risk predictions. It verifies that each module works properly, including symptom collection, data preprocessing, risk analysis, recommendation generation, and alert handling.

The tests also check overall system performance in real use conditions by evaluating prediction accuracy, consistency, and stability across different patient profiles and scenarios.

Safety validation ensures that all recommendations follow medical guidelines and do not generate harmful or misleading advice. The system is also tested for compliance with privacy and data handling standards.

Finally, performance tests confirm that the system responds within acceptable time limits and operates efficiently in real-time without delays or failures.

2.3.2.2. Test environment and configuration

A simulation harness runs repeated test episodes using the structured patient dataset. Each episode starts with a patient profile containing symptoms, diet, lifestyle, and medication history, along with a computed vitamin risk target output from the decision module. The system then processes this input through the full pipeline, including preprocessing, risk prediction, and recommendation generation, and records the final outputs until convergence or completion criteria are met.

Two trained models are evaluated: a low-risk model for early or mild deficiency patterns and a high-risk model for more severe or long-term deficiency cases. Each model is tested independently under controlled conditions to ensure stability and consistency of predictions across different patient scenarios.

For reproducibility, random seeds are fixed for each test scenario, and dataset versions, model checkpoints, and configuration files (JSON/YAML) are stored with each run. Testing is executed on both development and deployment environments to evaluate response time, prediction stability, and system performance under real-world constraints.

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2.3.2.3. Datasets and scenarios

The scenario suite reflects realistic patient conditions and risk variations in vitamin deficiency detection. It includes early-stage cases with mild symptoms (fatigue, low mood, poor sleep), moderate cases with combined dietary and lifestyle gaps, and severe cases involving long-term medication use and multiple deficiency indicators. Additional scenarios simulate special conditions such as vegetarian or restricted diets, low sunlight exposure, chronic illness, and inconsistent symptom reporting.

Each scenario provides an initial patient profile along with contextual information such as age group, diet pattern, and medication history, allowing the system to generate a baseline risk assessment and apply the decision logic for vitamin B12, D, and folate deficiencies.

Scenario coverage spans both typical and edge-case health conditions, ensuring the model is tested across a wide range of symptom combinations, dietary patterns, and medication interactions. This helps evaluate generalization, stability, and early detection capability under realistic and diverse patient inputs.

2.3.2.4. Metrics and acceptance criteria

System effectiveness is evaluated using prediction accuracy, early detection capability, and recommendation quality. The primary metric is correct classification of vitamin deficiency risk levels (low, moderate, high) for Vitamin B12, D, and folate, measured against validated or labeled reference outcomes.

Performance is assessed using prediction consistency over time, ensuring that repeated inputs from the same patient produce stable risk outputs. The system must also maintain low false-negative rates, especially for high-risk cases, to ensure early detection is not missed.

Recommendation quality is evaluated based on how accurately dietary, lifestyle, and supplementation suggestions align with the identified deficiency risks. Acceptance requires that recommendations remain medically consistent and relevant to the predicted condition.

Real-time performance is measured by system response time from data input to final risk output, with the requirement that predictions are generated within acceptable limits for interactive use.

Safety criteria require that all outputs remain within medically safe guidelines, avoiding overestimation of risk or harmful supplementation suggestions. All results are reported using average performance and percentile-based evaluation across multiple patient scenarios to ensure reliability and consistency.

2.3.2.5. Module (unit) testing

Symptom data collection module. Tests verify that patient inputs (symptoms, diet, lifestyle, and medication history) are correctly captured and structured into a consistent format. The module ensures all required fields are stored properly, severity scales are applied correctly, and missing or incomplete inputs are flagged for fallback handling without breaking the pipeline.

Data preprocessing module. Unit tests confirm that raw patient data is accurately normalized into standardized feature vectors. This includes correct scaling of symptom severity, encoding of dietary intake, transformation of medication history, and handling of missing values. Outputs are validated for consistency and correct dimensional structure.

Risk prediction module (AI model). Tests validate that the model correctly outputs vitamin deficiency risk scores and classifications for B12, D, and folate. The system is checked for stable predictions under repeated inputs, correct ranking of risk severity, and sensitivity to symptom–diet–medication combinations.

Decision function module. Tests ensure that the system generates a valid risk threshold decision (low, moderate, high) based on patient context. It verifies correct mapping of combined features into risk categories and confirms fallback behavior when incomplete data is provided. Medical logic consistency is also checked (e.g., known symptom–deficiency relationships).

Progression estimation module. This module is tested to confirm correct estimation of deficiency progression based on symptom duration and medication history. Outputs are validated for logical consistency and bounded within realistic clinical ranges.

Recommendation generation module. Unit tests verify that dietary, lifestyle, and supplementation recommendations are correctly generated based on predicted risk levels. The module is checked for safety compliance, ensuring no conflicting or harmful recommendations are produced.

Alert and notification module. Tests confirm that high-risk cases trigger appropriate alerts and notifications. The system is validated to ensure correct threshold detection and timely response without duplicate or missed alerts.

Explainability module. Tests verify that the system correctly traces outputs back to input features, ensuring that symptom, diet, and medication contributions are clearly identifiable in the final explanation.

End-to-end consistency checks. Final unit-level validation ensures that all modules work together without mismatch in data formats, that outputs remain stable across repeated runs, and that the full pipeline produces consistent and interpretable results from input to final recommendation.

2.3.2.6. Risk-model verification

The risk prediction model is verified by comparing predicted vitamin deficiency risk levels with ground-truth or validated patient outcomes across multiple scenarios. Outputs for Vitamin B12, D, and folate are evaluated against expected clinical patterns derived from symptoms, diet, and medication profiles.

Bias and variance are measured across different patient groups and conditions to ensure stable performance and avoid overfitting to specific symptom patterns. Sensitivity analysis is conducted to observe how changes in symptoms, dietary intake, and medication history affect risk scores.

Acceptance is satisfied when the median prediction error remains within an acceptable threshold ($\leq 20\%$) and when the error distribution remains consistent across low, moderate, and high-risk cases. The system is also expected to maintain balanced performance across all vitamin categories without favoring any single class.

2.3.2.7. Multi-agent selection tests

Multi-agent selection tests verify that the system correctly chooses between the low-risk and high-risk prediction models based on patient condition severity. Test cases simulate early-stage, moderate, and severe deficiency profiles to ensure the correct agent is activated for each scenario.

The **Agent-Low** model is expected to handle mild symptom patterns such as slight fatigue or minor dietary gaps, while the **Agent-High** model is expected to respond to complex cases involving multiple symptoms and long-term medication interactions.

Tests confirm that agent switching is consistent, deterministic, and based only on current patient input rather than previous states. Incorrect agent selection, delayed switching, or overlapping activation is treated as a failure case.

Acceptance is satisfied when the system consistently selects the appropriate agent for each risk level with high accuracy across all test scenarios, ensuring stable and reliable deficiency prediction behavior.

2.3.2.8. Logging, observability, and traceability

Each system run records structured outputs for full traceability of the prediction pipeline. For every patient update, the system logs timestamp, symptom inputs, preprocessing results, predicted vitamin risk scores, selected risk category, generated recommendations, and any alert events triggered.

The logging layer captures key metrics such as prediction confidence, risk level per vitamin (B12, D, folate), recommendation set generated, and system response time. Any fallback or rule-based inference events are also recorded with corresponding reasons for activation.

These logs enable full reconstruction of each decision path: raw patient input → processed features → risk prediction → decision function output → recommendation generation → final system output. Optional diagnostic artifacts include symptom trend summaries, risk evolution over time, and contribution analysis showing how symptoms, diet, and medication influenced the final prediction.

This observability framework ensures transparency, supports debugging, and allows healthcare professionals to review and validate system decisions for clinical reliability.

14.Results & Discussion

14.1. Results

This section presents quantitative evidence that the proposed system effectively identifies early vitamin deficiency risks from patient-reported symptoms, diet, lifestyle, and medication history. Results are obtained from a standardized evaluation pipeline using the structured patient dataset (real + synthetic profiles) and the trained multi-agent risk prediction models..

Evaluation metrics

System performance is evaluated using clinically relevant indicators:

- **Prediction accuracy (%)**: correctness of Vitamin B12, D, and folate risk classification (low, moderate, high).
- **Risk detection sensitivity (%)**: ability to correctly identify true deficiency cases, especially early-stage.
- **False negative rate (%)**: missed high-risk cases (critical safety metric).
- **Risk score error**: deviation between predicted risk score and reference label.
- **Response consistency**: stability of predictions across repeated inputs for the same patient.
- **Recommendation relevance score**: alignment of generated advice with detected deficiency type.

All metrics are computed per patient profile and summarized across different health scenarios including diet-restricted, medication-heavy, and lifestyle-imbalanced cases.

Overall performance

In later evaluation runs, the system shows stable and accurate risk detection across most patient profiles. High-risk cases are consistently identified, and early-stage deficiencies are detected with improved sensitivity.

Representative evaluation outcomes show:

- Mild cases correctly classified as low risk with stable outputs
- Moderate deficiency cases accurately mapped to medium risk levels
- Severe deficiency profiles correctly flagged as high risk with alerts triggered

Occasional misclassifications occur in borderline cases with incomplete dietary or medication data, but these are corrected in subsequent updates through longitudinal tracking. Overall, the system demonstrates strong consistency and improved detection reliability over repeated evaluations.

Progressive design evaluation

A series of design improvements led to steady performance gains:

- **Structured feature representation:** Converting raw symptoms into normalized feature vectors improved prediction stability and reduced noise sensitivity.
- **Multi-source integration:** Combining symptoms, diet, and medication history significantly improved detection accuracy compared to symptom-only models.
- **Multi-agent modeling:** Separating low-risk and high-risk models improved sensitivity for both early and severe deficiency cases.
- **Progression tracking:** Incorporating temporal symptom trends reduced late-stage misclassification and improved early warning capability.
- **Recommendation filtering:** Adding safety constraints improved relevance and eliminated unsafe or contradictory suggestions.

Together, these improvements resulted in more consistent predictions, better early detection performance, and higher clinical reliability.

Learning and prediction behavior

The system shows a clear improvement in risk prediction stability as more patient data is accumulated. Early inputs may produce uncertain classifications, but accuracy improves significantly once symptom history and medication context are integrated.

The multi-agent structure ensures smoother transitions between risk levels, reducing abrupt or inconsistent changes in classification. High-risk cases remain consistently prioritized, ensuring early warning effectiveness is maintained.

Risk detection and safety performance

The model successfully identifies key deficiencies patterns such as:

- Chronic fatigue + poor diet → Vitamin B12 risk
- Low sunlight exposure → Vitamin D risk
- Long-term medication use (e.g., absorption-affecting drugs) → increased deficiency likelihood

Safety constraints ensure that no excessive or harmful supplementation advice is generated. All high-risk alerts are aligned with medically reasonable thresholds.

Longitudinal stability and tracking

Over repeated patient updates, the system maintains stable risk trends and improves prediction accuracy through accumulated history. Gradual symptom progression is correctly reflected in increasing risk scores, while improving cases show corresponding risk reduction.

This confirms that the system supports continuous monitoring rather than one-time prediction, making it suitable for long-term health tracking.

Summary

- Structured symptom–diet–medication modeling significantly improves early detection accuracy
 - Multi-agent design enhances sensitivity across mild and severe deficiency cases
 - Longitudinal tracking improves stability and reduces misclassification over time
 - Safety constraints ensure medically consistent and non-harmful recommendations
 - The system reliably supports early warning of Vitamin B12, D, and folate deficiencies
- These results indicate that the controller meets the central objective: **quickly moving the cabin toward the passenger-comfortable target temperature** while maintaining spatial uniformity and respecting practical HVAC constraints.

Overall, the results confirm that the system achieves its goal of early, accurate, and clinically meaningful vitamin deficiency risk detection with safe and personalized recommendations.

14.2. Research Findings

This section summarizes the key outcomes, patterns, and practical implications of the proposed early vitamin deficiency warning system. Findings cover prediction effectiveness, risk modeling behavior, decision logic, recommendation quality, robustness to missing data, safety compliance, and real-world applicability, aligned with the main objective of early detection and prevention of Vitamin B12, D, and folate deficiencies.

1) Overall control effectiveness

Across the full evaluation dataset, the system reliably identifies vitamin deficiency risk levels from patient-reported symptoms, diet patterns, lifestyle behavior, and medication history. In later evaluation stages, the model frequently achieves correct classification for Vitamin B12, D, and folate risk levels, with stable performance in both moderate and high-risk cases.

Representative outputs show consistent detection of low-risk profiles for healthy or balanced users, correct identification of moderate-risk cases with early warning signs, and accurate escalation of high-risk profiles where multiple deficiency indicators are present. In several cases, the system produces highly confident predictions when symptom clusters strongly align with known deficiency patterns (e.g., fatigue + poor diet + long-term medication use).

Isolated misclassifications occur in borderline or low-information cases where patient inputs are incomplete or contradictory (e.g., missing diet logs or vague symptom descriptions). However, these errors are not persistent and are typically corrected as additional data becomes available through longitudinal updates, indicating overall stability rather than model brittleness.

Interpretation. The system consistently maps multi-source patient data into meaningful vitamin deficiency risk categories while maintaining low false-negative behavior for high-risk cases. It meets its primary objective of early detection by reliably flagging potential deficiencies and maintaining stable performance across diverse patient profiles, even under imperfect or partial data conditions.

2) Action space, reward shaping, and learning dynamics

Three design choices significantly improved the performance of the vitamin deficiency risk prediction system:

- **Discrete risk classification space (Low / Moderate / High, 3 classes per vitamin).** Early experiments using overly granular or continuous risk scoring produced unstable

learning and inconsistent predictions. Constraining outputs into medically meaningful categories stabilized training and improved classification accuracy, especially for Vitamin B12, D, and folate risk levels.

- **Feature-level reward shaping.** Replacing a single global prediction loss with structured feature-aware shaping (bonus for correct early detection, stronger penalty for missed high-risk cases, and weighted error for borderline cases) allowed the model to better identify which symptoms, dietary patterns, and medication factors were most influential. This improved sensitivity to true deficiency signals and reduced false negatives in high-risk profiles.

- **Dual-agent specialization.** Using separate models for low-risk/early-stage cases and high-risk/complex cases improved stability across different patient conditions. The low-risk agent handles mild symptom patterns and preventive detection, while the high-risk agent focuses on multi-factor deficiency cases involving long-term medication and severe symptom combinations. This separation reduced prediction variance and improved overall reliability.

Interpretation. Learning was most effective when the problem structure was made explicit through clinically meaningful classes, feature-level feedback for symptom importance, and specialized models for different severity ranges.

3) Grid-to-zone actuation feasibility

Although the system predicts detailed risk outputs at a feature level, real healthcare applications cannot act on hundreds of micro-level signals independently. The decision-to-action translation—aggregating symptom, diet, and medication-based risk signals into a single patient-level risk output (Low / Moderate / High)—proved effective. Final outputs were stable, clinically interpretable, and compatible with real-world screening workflows, while still preserving the benefit of fine-grained feature analysis during inference. The model’s internal feature attention ensures that critical signals (e.g., chronic fatigue, poor diet, long-term medication use) are naturally weighted more heavily in the final aggregated decision. The simplifying assumption of a single consolidated risk label per vitamin is sufficient for deployment in early warning systems while maintaining clinical usability.

Interpretation. The approach bridges high-dimensional patient data with practical clinical decision-making; detailed symptom-level intelligence is not lost when mapped into a single, actionable risk classification.

4) Generalization from simulated data Practical deployment feasibility

Although the system predicts risk at a detailed feature level (symptoms, diet, and medication history), real healthcare deployment cannot rely on highly granular, isolated signals for decision-making. The feature-to-decision translation—compressing multi-dimensional patient inputs into a unified vitamin risk output (Low, Moderate, High)—proved effective in maintaining both clinical interpretability and system stability. This

aggregation preserves the influence of key indicators such as fatigue, poor dietary intake, and long-term medication use, while ensuring outputs remain actionable in real-world screening settings.

The model trained on a diverse simulated and structured dataset generalizes well across different patient profiles, including varying diets, age groups, and lifestyle conditions. Even though the underlying inputs are high-dimensional, the final risk classification remains consistent and robust under different combinations of symptoms and missing data scenarios.

The simplifying assumption of a single consolidated risk label per vitamin is sufficient for practical deployment in early warning systems, as it balances predictive depth with clinical usability and decision clarity.

Interpretation. The approach bridges complex patient-level data with practical healthcare decision-making; detailed symptom and history-level intelligence is not lost, but effectively summarized into a single, clinically meaningful risk output.

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5) Quantitative outcome summary

- **Prediction performance.** In later evaluation stages, the system frequently achieves correct classification across Vitamin B12, D, and folate risk levels, with most patient profiles consistently mapped to the correct Low / Moderate / High category. Early-stage cases show improved detection sensitivity, while high-risk cases are reliably flagged with minimal missed detections. A small number of borderline or incomplete-input cases show inconsistent classification, but these are not persistent across longitudinal updates.
- **Risk consistency.** At convergence, the system maintains stable risk outputs across repeated patient inputs, with most predictions remaining within a consistent risk band. The deviation in risk scores reduces as more symptom history, dietary patterns, and medication context are included, improving overall stability.
- **Decision stability.** Risk classifications evolve smoothly over time without abrupt or oscillatory changes between severity levels. This ensures that patient monitoring reflects gradual health progression rather than unstable fluctuations.
- **Recommendation accuracy.** Generated dietary, lifestyle, and supplementation guidance remains aligned with the predicted deficiency type, with improved consistency as risk confidence increases. Early and preventive recommendations remain stable and medically relevant across test cases.

Interpretation. The quantitative profile confirms that the system achieves its objective of stable, early, and clinically consistent vitamin deficiency risk detection, suitable for real-world screening and continuous monitoring applications.

6) Decision function and comfort constraints

The risk level produced by the decision function—based on clinically guided symptom interpretation and constrained by safe medical thresholds for Vitamin B12, D, and folate—was consistently respected by the prediction pipeline. The safety guardrails prevented overestimation of risk in low-information or ambiguous cases, ensuring that output remained clinically reasonable and non-alarming. The system converged without triggering invalid or unsafe recommendation states.

Interpretation. The combination of a medically informed decision function and a safety-constrained prediction pipeline provide a reliable and defensible framework for early vitamin deficiency detection, suitable for real-world healthcare screening and monitoring applications.

7) Limitations and boundary conditions.

- **Occasional borderline cases.** A small number of patient profiles show unstable or less confident predictions, especially when symptom data is incomplete, inconsistent, or self-reported with low accuracy. These cases typically involve overlapping symptoms between different vitamin deficiencies, but performance stabilizes once additional context (diet or medication history) is available.
- **Simulation-to-real gap.** Parts of the system are trained and validated using structured or synthetic patient profiles. Real-world clinical data may include more noise, missing values, and unstructured reporting patterns, requiring further validation in real healthcare environments to confirm robustness.
- **Feature granularity vs. simplicity.** The system balances detailed feature analysis (symptoms, diet, medication history) with a simplified output (Low / Moderate / High risk). While this improves usability, it may reduce granularity in borderline clinical interpretations and could be refined with more advanced scoring layers in future work.

Interpretation. These limitations do not reduce the core feasibility of the system; instead, they define clear directions for future improvement, including real-world clinical validation, improved data quality handling, and enhanced risk scoring refinement.

14.3. Discussion

This study demonstrates that treating patient health as a multi-factor risk field rather than a single symptom indicator significantly improves early vitamin deficiency detection. By integrating structured patient-reported data (symptoms, diet, lifestyle, and medication history), the system is able to generate more stable and clinically meaningful risk predictions for Vitamin B12, D, and folate deficiencies. Across diverse patient profiles, risk scores consistently converge toward correct classifications after model refinement, indicating that the system effectively captures underlying deficiency patterns rather than relying on isolated symptoms.

Several design choices were critical to this performance. Structured feature-level inputs allowed the model to identify which factors contribute most to deficiency risk. Discrete risk categories (Low, Moderate, High) stabilized prediction behavior compared to continuous scoring approaches, which previously showed inconsistent outputs. Incorporating feature-aware weighting improved sensitivity to early warning signs and reduced missed high-risk cases. The dual-agent strategy further improved robustness by separating early-stage detection from complex multi-factor cases involving long-term medication use and combined symptom patterns.

The system is also designed with clinical safety in mind through medically guided decision constraints and bounded risk outputs. This ensures that predictions remain interpretable and suitable for real-world screening and patient monitoring.

Key limitations include reliance on structured and partially synthetic patient profiles, which may not fully capture real-world clinical variability such as incomplete reporting and irregular symptom descriptions. Additionally, simplifying output into three risk levels improves usability but reduces fine-grained clinical nuance in borderline cases.

Despite these constraints, the architecture is modular, scalable, and suitable for real-world deployment. It supports continuous monitoring, privacy-preserving local inference, and easy integration into healthcare decision-support systems. Future improvements include validation with real clinical datasets, enhanced handling of missing or noisy patient data, and expansion toward more detailed nutrient-level risk scoring and personalized intervention modeling.

Overall, the results indicate a system that is accurate, stable, and practical for early vitamin deficiency warning, with strong potential for clinical decision support and preventive healthcare applications.

15.Summary of Individual Contribution –Reshmi M G G K (IT22600448)

The contribution covers the complete development pipeline for an early vitamin deficiency warning system, from data preparation to deployable models and evaluation framework. Structured patient data was designed to capture symptoms, dietary patterns, lifestyle behavior, and medication history in a consistent format suitable for AI-based risk prediction.

A preprocessing specification was developed to convert raw patient inputs into normalized feature vectors with quality checks to handle missing or inconsistent data. A clinically guided decision function was implemented to classify vitamin deficiency risk levels (Low, Moderate, High) for Vitamin B12, D, and folate based on structured input data, ensuring alignment with safe medical interpretation standards.

A feature-aware learning framework was integrated to improve early detection sensitivity, including weighted learning for high-risk cases and structured handling of borderline conditions. A multi-agent architecture was introduced, separating low-risk and high-risk prediction models to improve stability and reduce variance across different patient profiles.

A practical recommendation layer was developed to generate personalized dietary, lifestyle, and supplementation guidance based on predicted risk levels, while maintaining safety constraints to avoid harmful or excessive recommendations.

A longitudinal tracking and monitoring mechanism was implemented to analyze symptom progression over time, enabling continuous risk updates and improved early warning capability. A simulation and evaluation framework was also developed to test system performance across diverse synthetic and structured patient scenarios, supporting validation of accuracy, consistency, and robustness.

The work produced versioned model artifacts, configuration files, and documentation suitable for integration into healthcare decision-support systems, with clear pathways for future clinical validation and real-world deployment.

16. Conclusion

This work set out to enable early detection of vitamin deficiency risks in patients while keeping predictions stable, interpretable, and suitable for real-world healthcare use. The problem was reframed from isolated symptom checking to a structured, multi-factor decision system that analyzes symptoms, diet patterns, lifestyle factors, and medication history together to estimate risk levels for Vitamin B12, D, and folate deficiencies. The resulting system integrates four main components: (1) structured patient data representation, (2) a clinically guided decision function for risk classification, (3) a feature-aware learning framework for early detection, and (4) a reinforcement-style multi-agent prediction strategy that separates low-risk and high-risk cases for improved stability and accuracy. A longitudinal tracking layer further enables continuous monitoring of patient health over time.

Because real labeled datasets for combined symptoms, diet, and medication-based deficiency prediction are limited, a structured and simulated patient dataset was used to cover a wide range of realistic health scenarios. These included dietary restrictions, long-term medication usage, lifestyle imbalances, and varying symptom severity levels. The dataset was curated to ensure consistency and reviewed for medical plausibility, enabling repeatable training and evaluation while encouraging generalization across diverse patient profiles.

Methodologically, three design choices were critical. First, structuring outputs into discrete clinical risk categories (Low, Moderate, High) improved interpretability and reduced unstable predictions seen in continuous scoring approaches. Second, introducing feature-level weighting allowed the system to better focus on key medical indicators such as fatigue, dietary deficiency patterns, and medication-related absorption issues, improving early detection sensitivity. Third, the multi-agent architecture separated simple early-stage cases from complex high-risk profiles, reducing variance and improving reliability across different patient conditions. Together with longitudinal tracking, these choices enabled consistent and clinically meaningful risk predictions.

The results demonstrate that the system meets its primary objective: early and reliable identification of potential vitamin deficiencies with stable and interpretable outputs. High-risk cases are consistently flagged for timely intervention, while moderate and low-risk cases remain stable across repeated evaluations. The recommendation module provides targeted dietary, lifestyle, and supplementation guidance aligned with predicted deficiency types, supporting preventive healthcare decisions.

From an engineering perspective, the system is modular and deployable. Inputs, preprocessing, prediction, and recommendation modules are clearly defined, allowing easy integration into healthcare decision-support platforms. Structured logging and monitoring support traceability, validation, and future model improvement. The system also includes safe fallback behavior when input data is incomplete or uncertain, ensuring continuous operation without producing unsafe outputs.

However, several limitations must be acknowledged. The system relies partly on structured or simulated patient data, which may not fully capture the variability and noise of real clinical environments. Borderline cases with incomplete or inconsistent inputs may still produce uncertainty in classification. Additionally, reducing outputs to three risk levels simplifies interpretation but may limit fine-grained clinical detail.

in complex cases.

These limitations highlight clear directions for future improvement. Real-world clinical validation using hospital or population-level datasets is essential to further improve robustness. Enhanced handling of missing or noisy patient inputs, along with more advanced probabilistic risk scoring, could improve accuracy in borderline cases. Future extensions may also include personalized health recommendations, integration of wearable sensor data, and real-time adaptive learning for continuous improvement. In summary, this research presents a structured, interpretable, and scalable approach to early vitamin deficiency risk detection. By combining multi-source patient data, feature-aware learning, and longitudinal monitoring, the system provides a strong foundation for preventive healthcare and clinical decision support, with clear potential for real-world deployment and future enhancement.

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